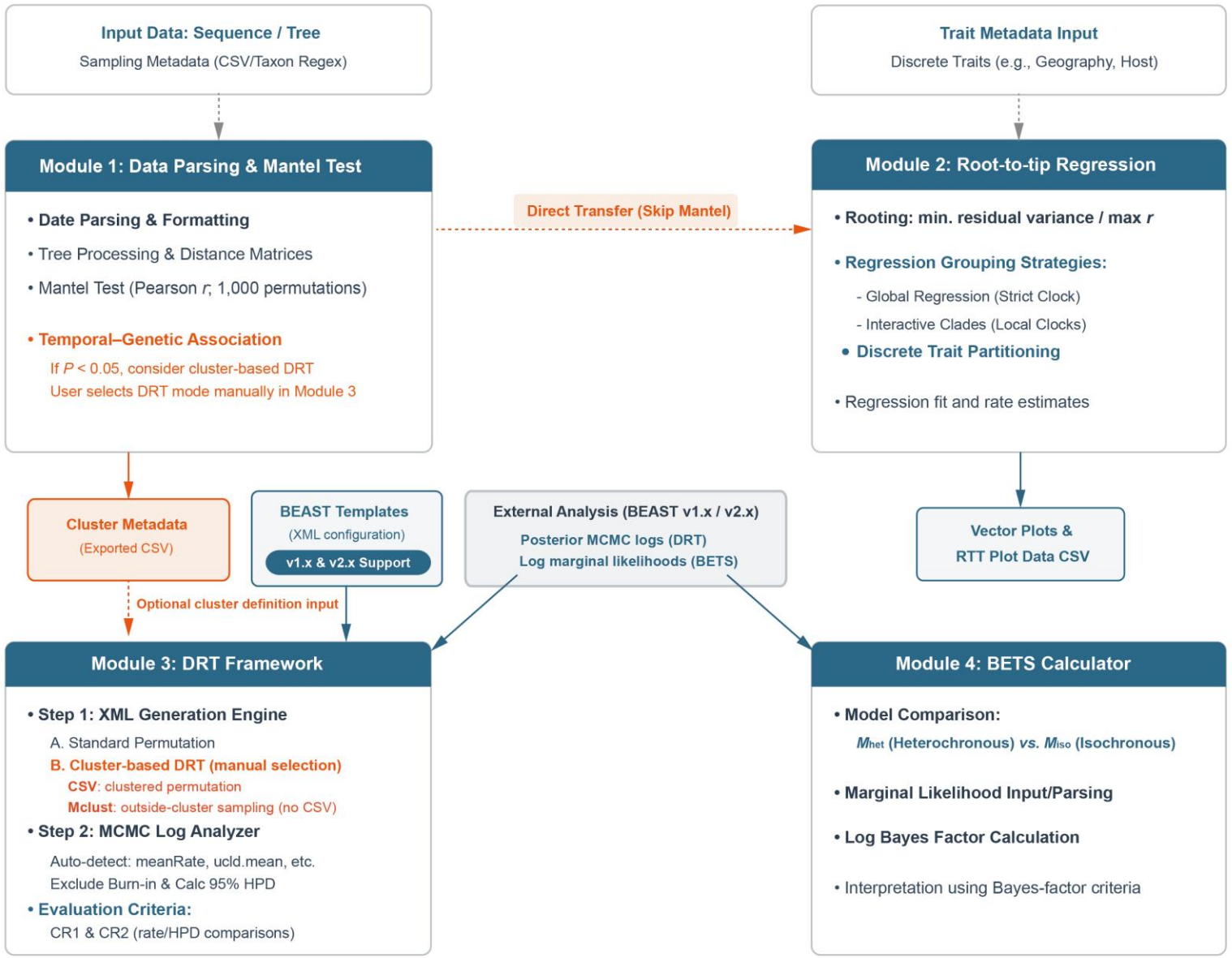
A quick guide to EvoTempo



Oct, 2026

Presented by Enhui Li et al.

I EvoTempo workflow





EvoTempo

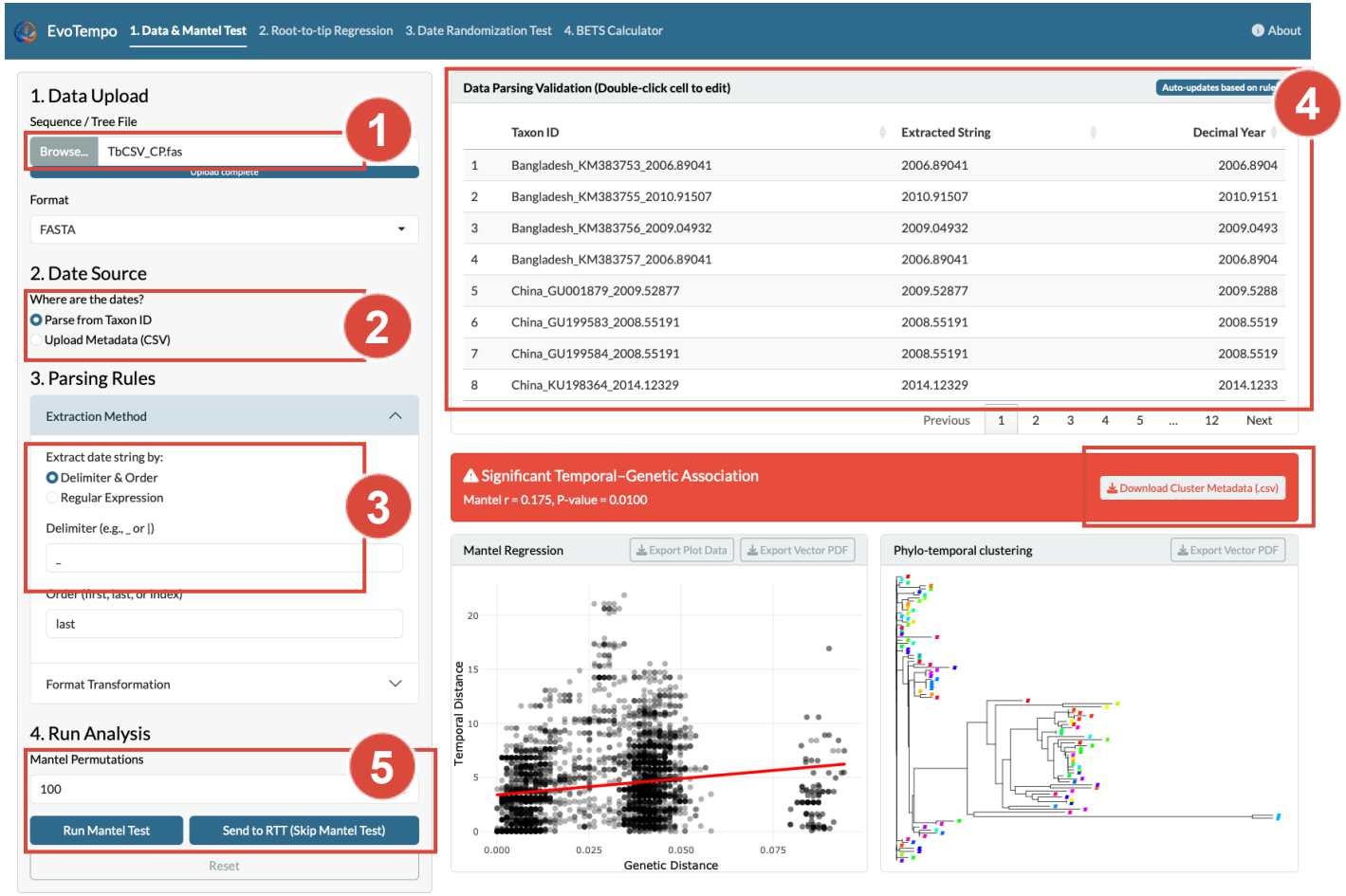
- **Module 1 → Module 2**: Click ***Run Mantel Test*** or ***Send to RTT (Skip Mantel Test).*** **Module 2** has no separate tree/date upload.
- **Module 1 → Module 3**: Export the cluster CSV, then upload it in Module 3.
- **DRT mode**: Select ***Enable Cluster-based DRT*** manually; the Mantel result does not change this setting.

II Input data, key configuration options, and outputs

Module	Required input(s)	Key configuration options	Main output(s)
Module 1: Data processing and Mantel test	Phylogenetic tree(.nwk, .tree, .nex) or sequence alignment (.fas, .fasta); optional sampling date metadata (.csv)	Delimiter or regex-based date parsing; interactive decimal-year table editing; Mantel permutation count (default: 1,000)	Parsed tree and date vector; temporal and genetic distance matrices; Mantel *r* and *P*-value; phylo-temporal cluster tree (PDF) and cluster metadata (.csv)
Module 2: Interactive RTT regression	Phylogenetic tree and corresponding sampling dates (passed from Module 1); optional discrete trait metadata (.csv)	Rooting strategy (nve-rms or max r); grouping mode (Global Clock, Interactive Clades / Local Clocks, or Discrete Trait Partitioning); permutation test (1,000 replicates)	Color-synchronized interactive phylogeny and RTT regression plot; group-specific Rate, *x*-intercepts, and R^2 summaries; grouped data (.csv) and vector plots (PDF)
Module 3: Date-randomization test (DRT)	BEAST v1.x/v2.x XML template; optional cluster metadata (.csv/.tsv); original and randomized MCMC .log files	Randomization mode (Standard vs. Cluster-based); clustering and randomization strategy (CSV-based clustered permutation vs. Mclust-based outside-cluster sampling); requested replicate count (5–20); Mclust model (E/V), maximum G and optional date rounding; burn-in percentage and rate parameter	ZIP archive of date-randomized BEAST XMLs and a randomization-strategy record; 95% HPD interval summaries; CR1 and CR2 validation cards; interactive and PDF 95% HPD forest plot
Module 4: Bayesian evaluation of temporal signal (BETS)	Marginal likelihood output logs (GSS, PS, or SS) or manual log-marginal likelihood values for competing models	Competing model specification (M_{het} vs. M_{iso}); automated log marginal likelihood text extraction or manual entry	Log marginal likelihood estimates; log Bayes Factor; Kass & Raftery evidence classification; dumbbell comparison plot (PDF)

Note: Optional inputs and conditional outputs (e.g., confounding cluster assignments in Module 1, trait metadata in Module 2, and cluster definitions in Module 3) are invoked depending on dataset structure and user-selected analytical modes. Detailed parameter specifications, file format templates, and statistical decision criteria are provided in the EvoTempo User Manual. The number of generated CSV-based replicates may be lower than requested when the available clustered permutations are limited. RTT, root-to-tip; tMRCA, time to the most recent common ancestor; DRT, date-randomization test; HPD, highest posterior density; BETS, Bayesian evaluation of temporal signal; MCMC, Markov chain Monte Carlo; GSS, generalized stepping-stone sampling; PS, path sampling; SS, stepping-stone sampling; M_{het}, heterochronous model; M_{iso}, isochronous model.

III Module Walkthrough | Module 1: Data & Mantel



Step-by-step operation

1. **Upload Data** – Upload a sequence alignment or tree. Check the detected file format and correct it if needed.
2. **Specify Date Source** –Choose Parse from Taxon ID or Upload Metadata (.CSV).
3. **Set Parsing Rules** – Choose the extraction method (delimiter or regular expression) and date format (decimal years or calendar dates).
4. **Verify Dates** – Review the preview table. Double-click Decimal Year cells to correct dates if needed.
5. **Run Mantel Test** – Set the number of permutations (default: 1,000) and click **Run Mantel Test**. If significant, click **Download Cluster Metadata (.csv)** to save the clusters. To skip testing, click **Send to RTT (Skip Mantel Test)** and continue in Module 2.

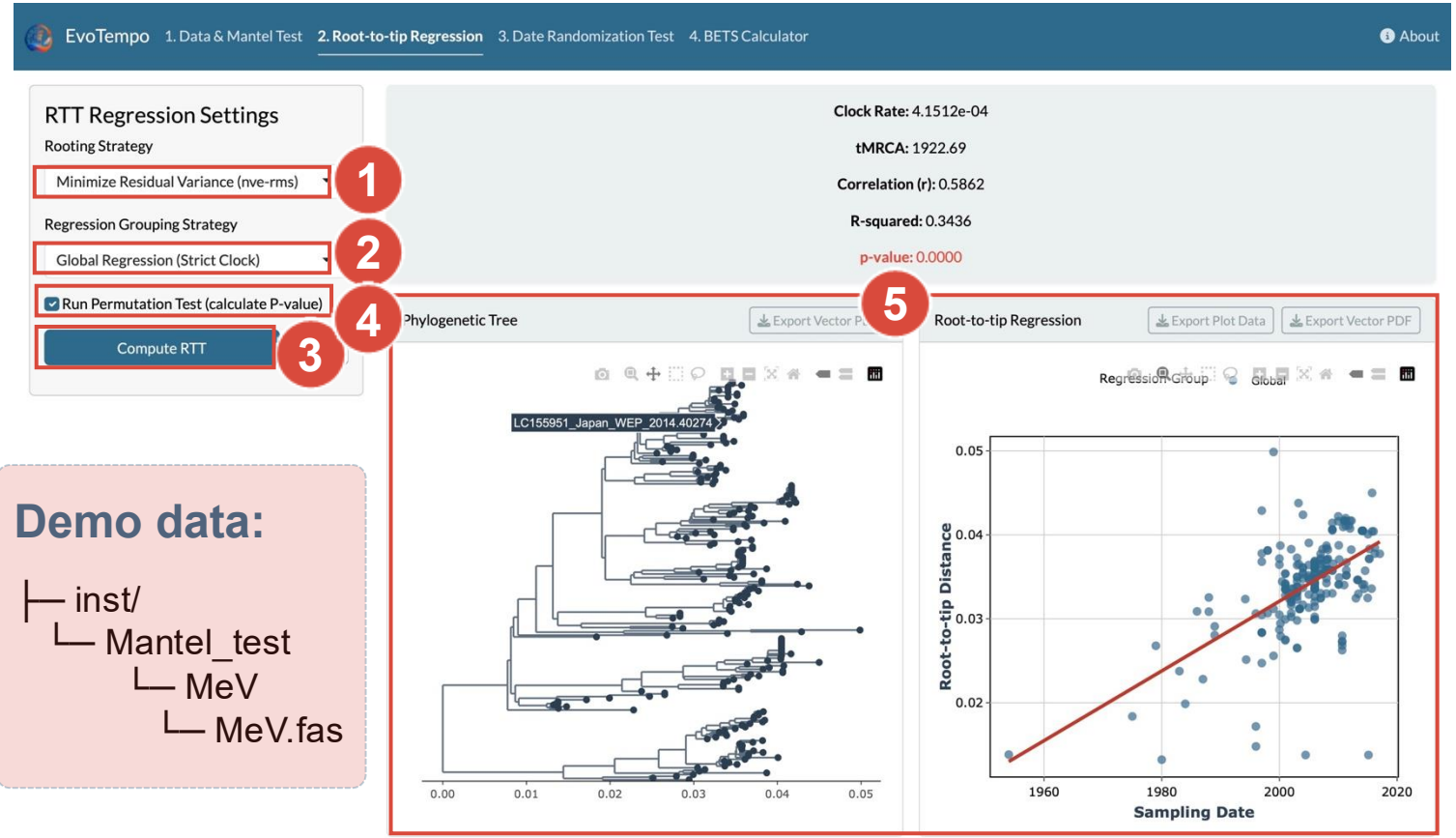
Results & Interpretation

- **P ≥ 0.05**: No significant temporal–genetic association detected; **P < 0.05**: Significant temporal–genetic association detected.

Consider the sampling design and these diagnostic results when choosing Standard DRT or Cluster-based DRT.

IV Module Walkthrough | Module 2 Root-to-tip Regression

Global Regression (Strict Clock)



Demo data:

```
├─ inst/
   └─ Mantel_test
      └─ MeV
         └─ MeV.fas
```

Step-by-step operation

1. **Select Rooting Strategy** – Choose Minimize Residual Variance (nve-rms) or Maximize Correlation (*r*) (see **Rooting Tip** below).
2. **Select Regression Grouping** – Choose Global Regression (Strict Clock).
3. **Compute RTT** – Click Compute RTT to optimize the root and display the regression.
4. **Optional Permutation Test** – Enable Run Permutation Test before computing RTT to assess the global correlation using 1,000 date shuffles. This exploratory test does not replace DRT or BETS.
5. **Export Results** – Export the tree and RTT plots (.PDF), and RTT plot data (.CSV).

Rooting Tip

- **nve-rms:** Minimizes RTT residual variance (default).
- ***r*** : Maximizes the Pearson correlation between sampling dates and root-to-tip distances.

IV Module Walkthrough | Module 2 Root-to-tip Regression

Interactive Clades (Local clocks)



Demo data:

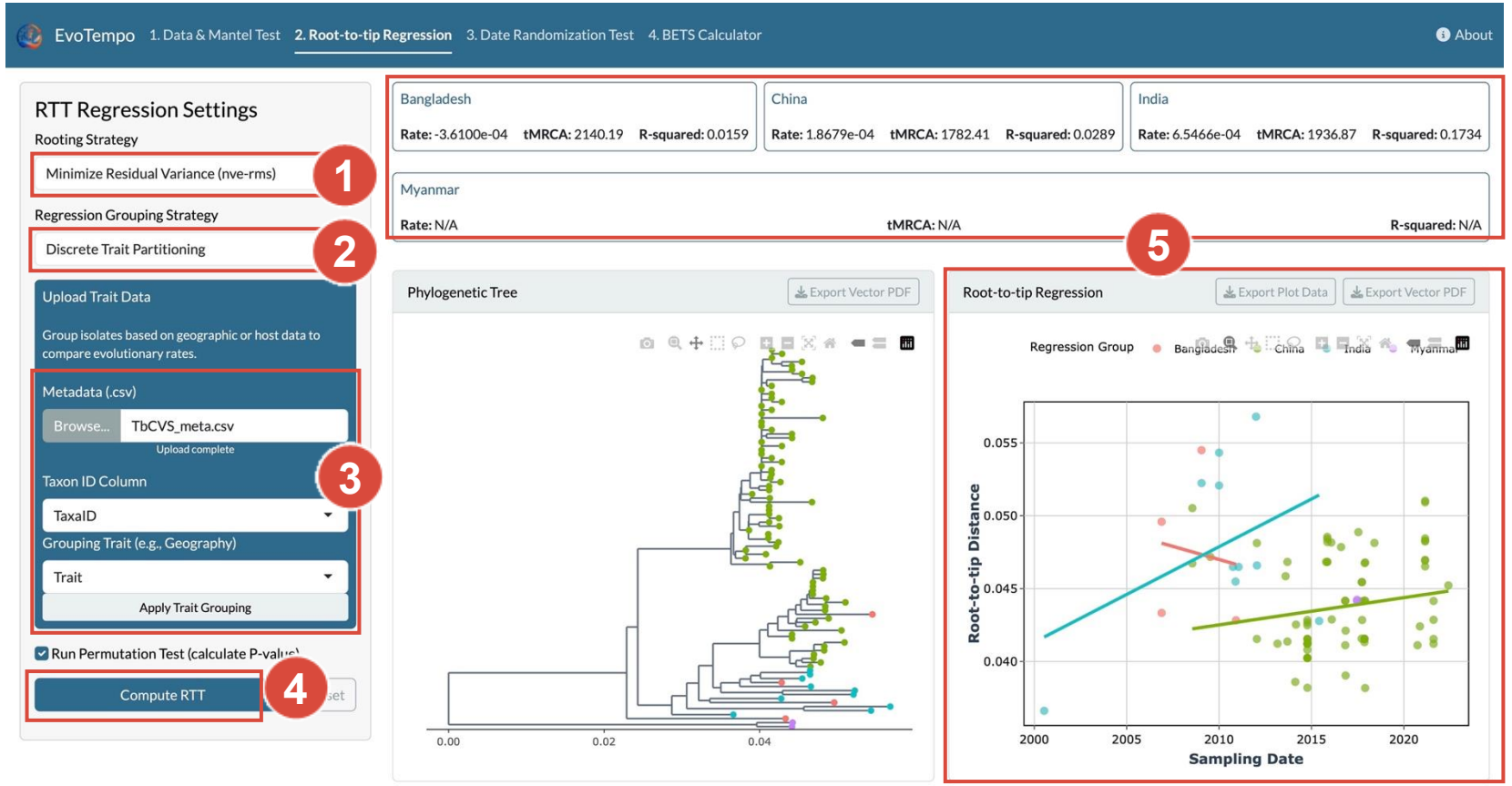
```
├ inst/
  └ Mantel_test
      └ TbCSV_CP.fas
```

Step-by-step operation

1. **Select Rooting Strategy –** Choose Minimize Residual Variance (nve-rms) or Maximize Correlation (r).
2. **Select Regression Grouping –** Choose Interactive Clades (Local Clocks).
3. **Assign Clades –** Click internal nodes to define color-coded clade groups. Use Clear Local Clocks to reset selections.
4. **Compute RTT –** Click Compute RTT to optimize the root and display clade-specific regressions.
5. **Review & Export –** Compare clade RTT slopes, then export plots (.PDF) and plot data (.CSV).

IV Module Walkthrough | Module 2 Root-to-tip Regression

Discrete Trait Partitioning



Demo data:

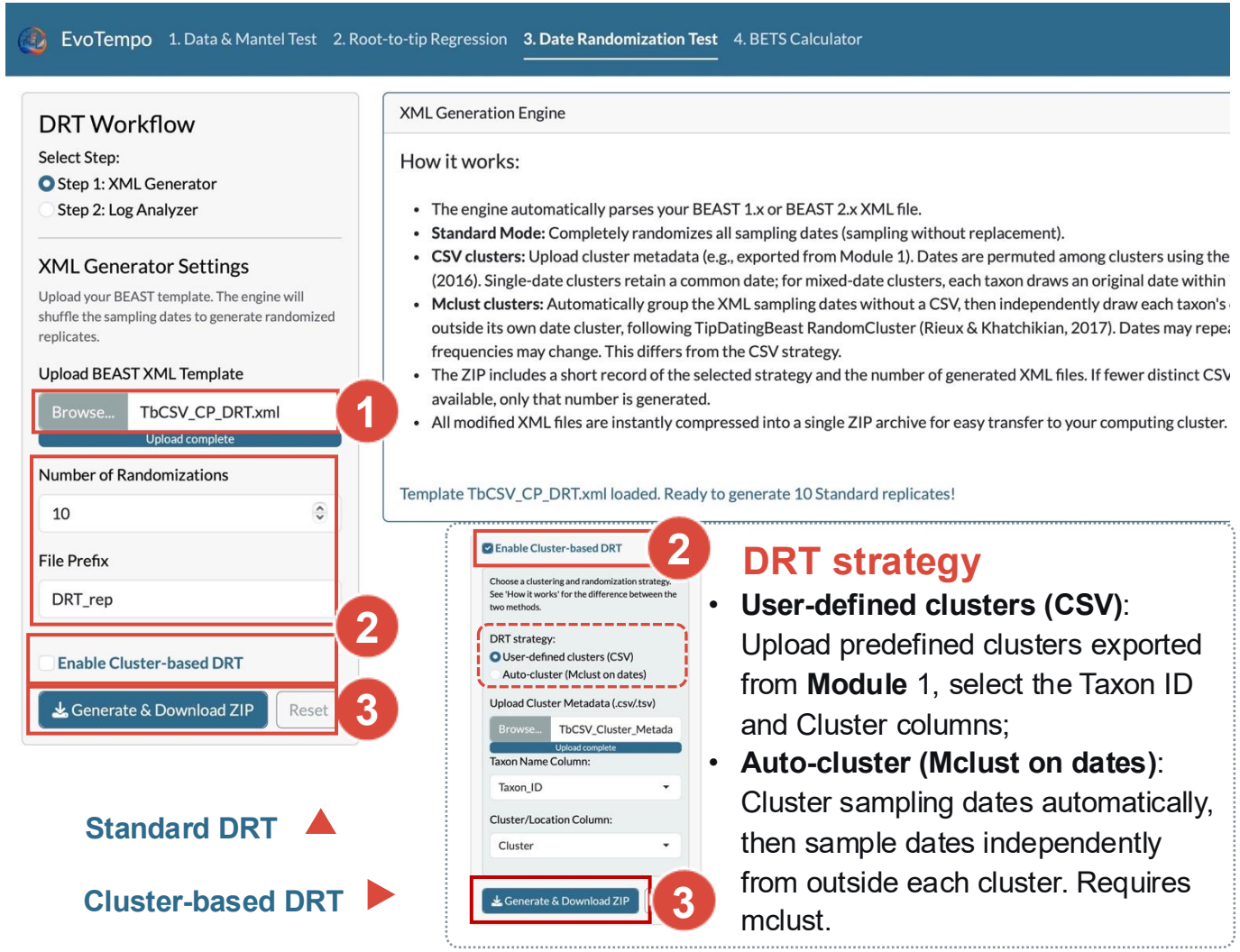
- inst/Mantel_test/TbCSV_CP.fas (*.fas)
- inst/RTT/TbCVS_meta.csv (metadata)

Step-by-step operation

1. **Select Rooting Strategy** – Choose Minimize Residual Variance (nve-rms) or Maximize Correlation (r).
2. **Select Regression Grouping** – Choose Discrete Trait Partitioning.
3. **Apply Trait Grouping** – Upload metadata (e.g., location or host). Select Taxon ID Column and Grouping Trait, then click Apply Trait Grouping.
4. **Compute RTT** – Click Compute RTT to optimize the root and display trait-group regressions.
5. **Review & Export** – Compare group RTT slopes, then export plots (.PDF) and plot data (.CSV).

V Module Walkthrough | Module 3 Date Randomization Test

XML Generator



DRT strategy

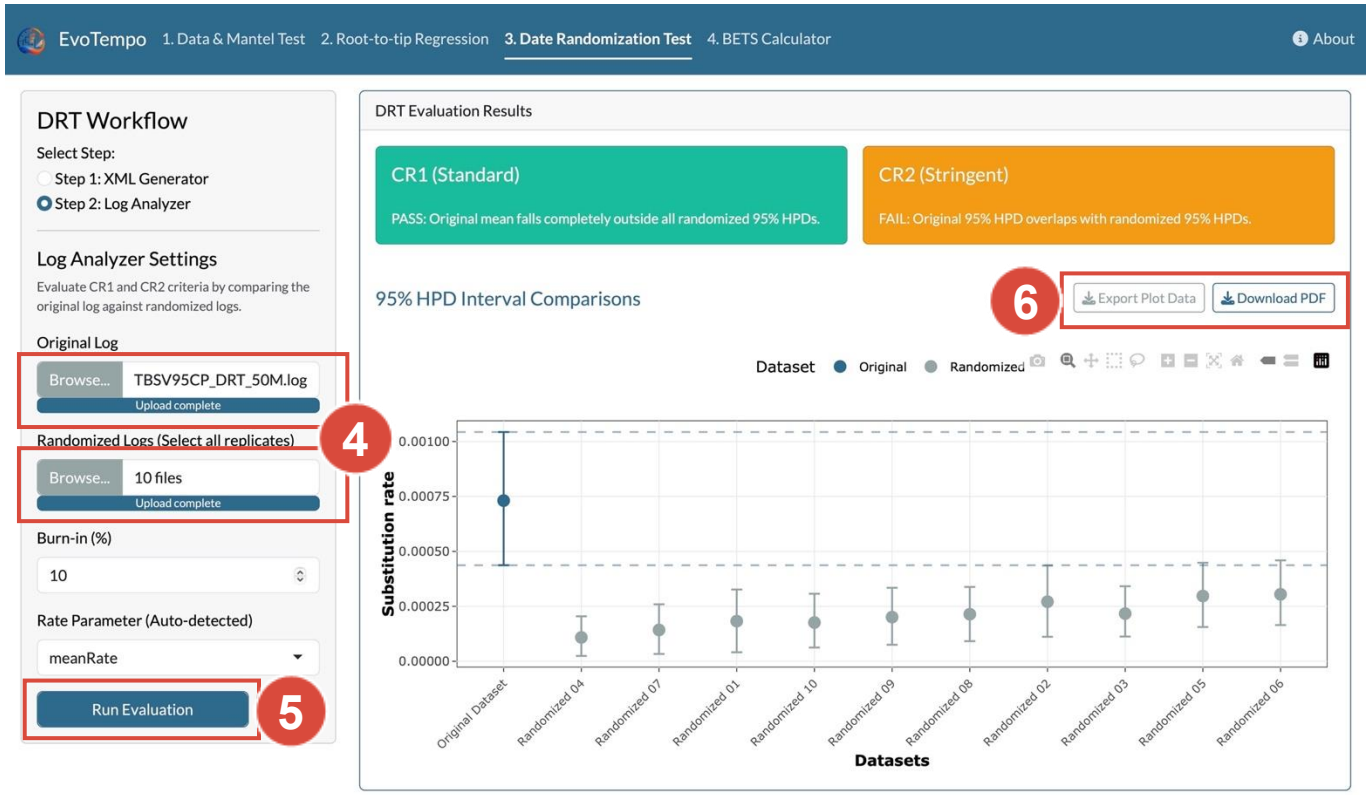
- **User-defined clusters (CSV)**: Upload predefined clusters exported from **Module** 1, select the Taxon ID and Cluster columns;
- **Auto-cluster (Mclust on dates)**: Cluster sampling dates automatically, then sample dates independently from outside each cluster. Requires mclust.

Step-by-step operation

1. **Upload & Configure** – Upload a BEAST XML template. Set the replicate count (default: 10) and file prefix (default: DRT_rep). Use distinct prefixes for Standard and Cluster-based DRT.
2. **Select DRT Mode** –
   - **Standard DRT**: Leave **Enable Cluster-based DRT** unchecked;
   - **Cluster-based DRT**: Check the option and choose a cluster source (see left).
3. **Generate & Download** – Click Generate & Download ZIP, then run the randomized XMLs in BEAST to obtain MCMC logs for the **Log Analyzer**.

V Module Walkthrough | Module 3 Date Randomization Test

Log Analyzer



Step-by-step operation

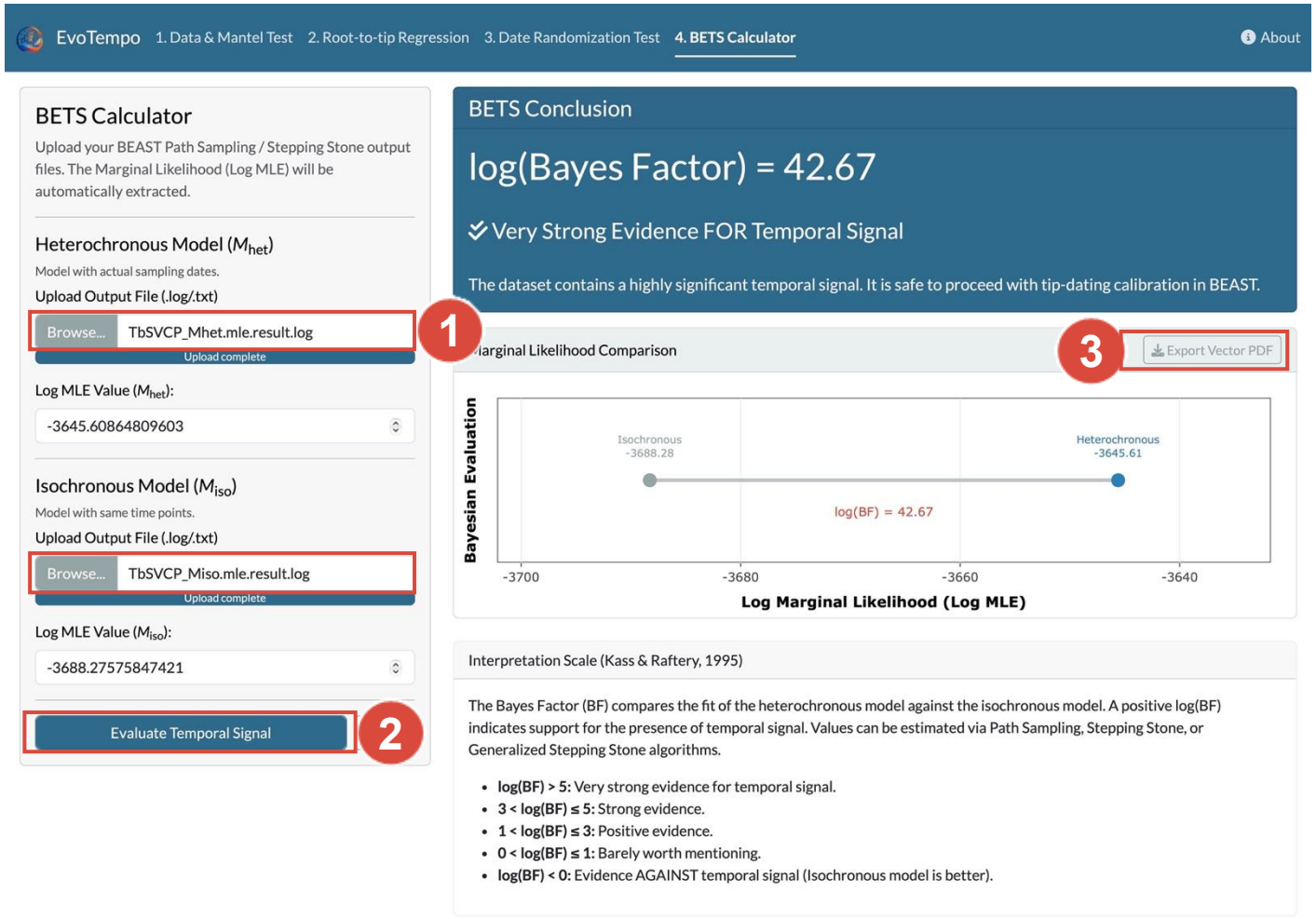
4. **Upload Logs** – Upload the original MCMC log and all randomized logs.
5. **Run Evaluation** – Check the rate parameter and burn-in, then click **Run Evaluation**.
6. **Export Results** – Export posterior summaries (.CSV) and the forest plot (.PDF).

Results & Interpretation

- **CR1**: Original posterior mean substitution rate lies outside every randomized 95% HPD.
- **CR2**: Original 95% HPD interval for the substitution rate does not overlap any randomized 95% HPD.

Touching boundaries fail either criterion.

VI Module Walkthrough | Module 4 BETS Calculator



Step-by-step operation

1. **Upload Data** – Upload M_{het} and M_{iso} log marginal-likelihood summaries from BEAST (e.g., GSS output), or enter values manually.
2. **Calculate & Interpret** – Click **Evaluate Temporal Signal** and review the evidence category.
3. **Export PDF** – Click Export Vector PDF to save the plot.

log(BF)	Interpretation
> 5	Very strong
> 3 to 5	Strong
> 1 to 3	Positive
> 0 to 1	Weak / inconclusive
= 0	Equal support
< 0	*M*iso preferred

Results & Interpretation

Positive log(BF) favors M_{het}; negative favors M_{iso}; zero indicates equal support.

VII Troubleshooting

- **Tree branches are missing from the RTT plot.**

  Check that the tree contains valid branch lengths. Re-export it in Newick or NEXUS format, reload it in **Module 1**, and rerun RTT.

- **DRT log evaluation fails.**

  Select the exact rate-parameter column present in the original log and **every randomized log**. Check the selection if automatic detection chooses an unintended parameter.

- **BETS extraction fails or returns an incorrect value.**

  Check the external analysis summary and manually enter the final **log marginal likelihood**. Use natural-log values, not maximum-likelihood estimates or ordinary MCMC trace values.

- **ggtree installation fails via install.packages().**

  Install it through Bioconductor: `BiocManager::install("ggtree")`.

- **Module 3 reports that mclust is required.**

  Run `install.packages("mclust")`, or choose **Upload Metadata CSV** to supply cluster definitions.

- **Mclust returns “only one cluster.”**

  Review the sampling dates and clustering settings, or upload justified cluster definitions. A single date cluster alone does not determine whether Standard DRT is appropriate.

- **Cluster metadata do not match the XML taxa.**

  Ensure Taxon IDs exactly match those in the XML, include every XML taxon, and select the correct **Taxon ID** and **Cluster** columns.

VIII Contact us

- **EvoTempo is an ongoing project**

  We welcome bug reports, feedback, and suggestions

- **Link**

  https://github.com/BioEasy/EvoTempo

- **Contact us**

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Try it yourself!